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Csql: Mapping Documents into Causal Databases *

A Preprint

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Abstract

We describe a novel system, Csql, that automatically converts a collection of unstructured text documents into an SQL-queryable *causal database* (CDB). A CDB differs from a traditional DB: it is designed to answer "why" questions via causal interventions and structured causal queries. Csql builds on our earlier system, Democritus, which converts documents into thousands of local causal models derived from causal discourse [Mahadevan, 2025b]. Unlike RAG-based systems or knowledge-graph–centric approaches, Csql supports causal analysis over document collections rather than purely associative retrieval. For example, given an article on the origins of human bipedal walking, Csql enables queries such as: "What are the strongest causal influences on bipedalism?" or "Which variables act as causal hubs with the largest downstream influence?"

Beyond single-document case studies, we show that Csql can also ingest *RAG/IE-compiled causal corpora* at scale by compiling the Testing Causal Claims (TCC) dataset of economics papers into a causal database containing 265,656 claim instances spanning 45,319 papers, 44 years, and 1,575 reported method strings, thereby enabling corpus-level causal queries and longitudinal analyses in SQL. Conceptually, Csql converts document collections into causally grounded relational databases, enabling causal analysis via standard SQL. In contrast to prior work that relies on hand-designed ontologies, fixed schemas, or domain-specific information extraction pipelines, Csql induces its schema directly from language (or from language-compiled causal artifacts). Viewed abstractly, Csql functions as a compiler from unstructured documents into a causal database equipped with a principled algebra of queries, and can be applied broadly across many domains ranging from business, economics, humanities, and science.

Keywords Causality · Natural Language · Databases · SQL · AI · Machine Learning

1 Introduction

We introduce Csql, a causal knowledge discovery system that automatically converts collections of documents into SQL-queryable *causal* databases (CDB). Unlike retrieval-augmented generation (RAG) systems or knowledge-graph–centric approaches, which yield traditional databases, Csql supports *causal analysis* over document collections rather than purely associative retrieval. Csql also differs fundamentally from traditional causal inference methods [Imbens and Rubin, 2015, Pearl, 2009], which typically focus on narrow, domain-specific studies grounded in numerical or experimental data. In contrast, Csql operates directly on large collections of unstructured text.

A central feature of Csql is that it induces its relational schema directly from language. Unlike prior approaches that rely on hand-designed ontologies, fixed schemas, or domain-specific information extraction pipelines, Csql automatically constructs a causally grounded database whose structure emerges from

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Property	RAG Systems	Knowledge Graphs	IE Pipelines	CSQL
Primary goal	Answer generation	Structured facts	Entity / relation extraction	Causal analysis
Input	Text chunks	Curated triples	Annotated text	Unstructured documents
Schema	None (implicit)	Predefined ontology	Fixed schema	**Induced from discourse**
Causality support	None	Implicit / informal	Pairwise only	**Weighted causal relations**
Uncertainty	None	Rare / ad hoc	None	**Aggregated model support**
Conflicting claims	Not supported	Manual resolution	Not supported	**Preserved and quantified**
Compositionality	Prompt-based	Limited graph traversal	Limited	**SQL joins and aggregations**
Counterfactuals	No	No	No	**Supported via causal structure**
Output format	Generated text	Graph database	Triples / tables	**Relational causal database**
Execution model	Neural inference	Symbolic querying	Batch extraction	**Deterministic SQL**

Table 1: Comparison of CSQL with RAG systems, knowledge graphs, and traditional information extraction (IE) pipelines. CSQL differs by compiling causal discourse into an SQL-queryable causal database with explicit uncertainty and support for causal analysis.

discourse itself. Viewed abstractly, CSQL functions as a compiler from unstructured documents into a causal database equipped with a principled algebra of queries. This design enables broad applicability across domains ranging from science and public policy to economics and history.

Table 1 contrasts traditional RAG or Knowledge Graph built databases with causal databases constructed by CSQL. CSQL builds directly on our previous work on DEMOCRITUS, which introduced an end-to-end system for discovering and evaluating causal structure from language. Given a document, DEMOCRITUS automatically constructs thousands of local causal models, evaluates them for plausibility, and aggregates them into a ranked causal analysis expressed in natural language. CSQL takes the output of DEMOCRITUS one step further by compiling these causal artifacts into an SQL-queryable database, thereby enabling systematic causal querying over entire document collections.

CSQL relies on recent advances in categorical causality [Mahadevan, 2025a, Fritz, 2020] and geometric deep learning [Fong et al., 2019, Mahadevan, 2024, Gavranović et al., 2024], together with large language models (LLMs) used as discourse compilers. LLMs can generate rich causal narratives—enumerating subtopics, posing causal questions, and articulating mechanisms across domains ranging from macroeconomics to neuroscience. However, LLM-based document summarization alone cannot produce a causally structured, queryable database. CSQL fills this gap by transforming causal discourse into an explicit relational representation.

For example, given a newspaper article—such as a recent *Washington Post* story on whether eating dark chocolate may elongate life,[2]—CSQL can construct a causal database that supports queries such as: "What are the strongest causal influences on dark chocolate consumption?" or "Which variables act as downstream causal hubs in this discourse?" More broadly, CSQL is designed to support probing, comparative analysis of causal claims not only within individual documents, but across an entire *frame of causal discourse* induced from a document collection.

A crucial design choice in CSQL is the construction of this frame of discourse. Rather than restricting analysis to the input documents alone, the system treats each document as a seed for a broader causal neighborhood. DEMOCRITUS first performs topic discovery over the input text and then expands these topics using a language model to construct a local discourse manifold. Causal claims are generated and evaluated relative to this expanded context. As a result, CSQL reasons not only about what a document explicitly states, but about how its claims are situated within a larger landscape of causal discourse.

2 The Origin of Bipedal Walking: A Running Example

We will illustrate the construction of CSQL through a running example based on The Washington Post article on the origins of bipedal walking in human ancestors, dating back to 7 million years ago (see Figure 1).[3]. The process of constructing a CSQL database involves first preprocessing the PDF document containing this article using our previous DEMOCRITUS [Mahadevan, 2025b] system. Democritus is a system for building large causal DAG models from carefully curated queries to a large language models (LLM) [OpenAI, 2025,

[2] `https://www.washingtonpost.com/wellness/2025/12/26/dark-chocolate-health-benefits`

[3] urlhttps://www.washingtonpost.com/science/2026/01/02/human-ancestor-biped/